



How can Continuous Performance Test help to assess inattention when mood and ADHD symptoms coexist?

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ABSTRACT

Depression and attention-deficit/hyperactivity disorder (ADHD) are prevalent, and often comorbid, disorders, with varying severity levels among patients. Inattention is a symptom present in both disorders, which often makes their differential diagnosis difficult in clinical practice (depression only versus comorbidity). This study aimed to investigate the influence of depressive symptoms on attention performance using one of the most common tasks in clinical practice, the continuous performance test (CPT). Ninety-three college students (60 men, 33 women) with a mean age of 24 years old were investigated with self-reports and semi-structured interviews for ADHD; the Beck Depression Inventory (BDI) was used for depression ratings. Attention measures were derived from the CPT. There was no correlation between depression and ADHD symptoms; in addition, depression was not correlated with any of the CPT scores; ADHD symptomatology was the only predictor of changes in those CPT variables (commission and omission errors and d prime). ADHD-associated impairment on the CPT was not augmented by the presence of depressive symptoms, making neuropsychological results on this test helpful for the differential diagnosis. When attention deficits are observed in individuals with mild or moderate depression, they are most likely not attributed to depression.

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1. Introduction

Depression and Attention-Deficit/Hyperactivity Disorder (ADHD) are highly prevalent disorders, often comorbid with each other (Kessler et al., 2005b), with varying severity in the population (Kooij et al., 2001; McGough et al., 2005; Murphy and Barkley, 1996). Both disorders are heritable, sharing familial pathophysiological and genetic risk factors (Biederman et al., 1993; Rommelse et al., 2008). The presence of comorbidities, like depression, may obscure an ADHD diagnosis (Adler, 2004). On the other hand, comorbid depression augments the impact of ADHD (Biederman, 2004; Secnik et al., 2005). Since both disorders comprise inattention as part of their clinical picture (American Psychiatric Association, 2013), differentiating among these diagnoses may be difficult in clinical practice. Another challenge seen by clinicians is to either make a differential diagnosis of depression or ADHD only versus that of depression with comorbid ADHD).

Severely depressed patients are seldomly referred to neuropsychological testing because depression can seriously compromise their performance, thus complicating interpretation of results (Lezak et al., 2012). Contrastingly, less severe cases are often referred to a neuropsychologist in order to help disentangle the cause of inattention re-

ported by the patient. This latter scenario demands a more thorough clinical investigation since both ADHD and depression may contribute to impairments on attentional tests. In fact, some authors (Groth-Marnat, 2000; Sweet et al., 1992) suggest caution when interpreting attentional deficits when mood symptoms are present.

For many decades, neuropsychologists have attempted to understand the neuropsychological dysfunction in depression. Beblo et al. (2011) have summarized several clinical and demographic factors, which potentially influence the profile and the severity of neuropsychological deficits of patients with mood disorders. Additionally, they have demonstrated that although those deficits are present mainly on attention, they are also manifest across a wide range of cognitive domains as executive functions and memory.

A large number of studies have been conducted in middle-aged and elderly depressed patients with mixed clinical variables that could account for the discrepant findings in the literature (Groth-Marnat, 2000). The type of mood disorder, its course, number of episodes and residual symptoms are some of such variables. Most studies commonly included only the more severe form of the disorder, Major Depressive Disorder (MDD) and also patients under medication, which might affect cognition, specially benzodiazepines and antidepressants with anticholinergic properties (Fossati et al., 2002). Thus, it is not clear if those results can be generalized to less severe depressive cases. Also, many studies comprised small samples and provided no data regarding depression severity or comorbidity (Westheide et al., 2007). Patients with Bipolar Disorder also have

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demonstrated impaired performance in Continuous Performance Tests (CPT) when compared to controls (Najt et al., 2005).

Fischer et al. (2005) demonstrated that depression did not contribute to impairments on the CPT of a clinical sample of young adults. In contrast, Larochette et al. (2011), reported that depressive symptoms significantly impaired the neurocognitive performance of young adults with ADHD comorbid with severe depression. However, this experiment attention was inferred through indirect indexes from the Wechsler Intelligence Scale and the Wechsler Memory Scale, rather than using attention tests. The use of tests tailored for attention assessment could document more specific deficits and contribute to a better understanding of the relation between mood symptoms and performance.

There are different CPT versions available for research use and relatively few studies using them on adult ADHD population (Mowinckel et al., 2015). Variability among different CPTs may affect their sensitivity and generalization of data on performance must be regarded with caution (Maoz et al., 2015). A meta-analysis including measures from different CPT versions demonstrated that the attentional and decision-making deficits of ADHD adults are of a similar magnitude when compared to controls (Mowinckel et al., 2015). Interestingly, such results contrast with those found in pediatric ADHD studies (Rossi et al., 2015) where their attentional profile is characterized by more variable performance, difficulty in rapidly adapting performance, slower processing speed, less efficient responses and reduced duration of the engagement of focus on tasks. Furthermore, ADHD patients with comorbid Borderline Personality Disorder (BPD) present changes in CPT response style with more widespread inhibition deficiencies (van Dijk et al., 2014).

Hill et al. (2008) investigated 161 adult patients of normal intelligence quotients (IQ) and found that self-reported depressive symptoms using the Beck Depression Inventory (BDI) minimally affected attention performance on the CPT (less than 1.5% of unique variance on all measures). This finding suggests that attention deficits within this population were most likely indicative of primary attention problems rather than functional impairment secondary to depression. In concert with this, some studies (Fischer et al., 2005; Morasco et al., 2006) also found that self-reported mood symptoms exerted minimal or equivocal influence on cognitive performance. Although there is evidence of a positive relationship between attention and depression (Watari et al., 2006) other studies failed to replicate such relationship (Rohling et al., 2002; Tsushima et al., 2005).

The aim of this study was to investigate how depressive and ADHD symptoms influenced changes on CPT performance. We have chosen to investigate college students because a relatively large number of college students (up to 8%) report clinically significant levels of ADHD symptomatology and at least 25% of those with disabilities receive an ADHD diagnosis (DuPaul et al., 2009). In addition, college students belong to an age group considered to be at risk for the emergence of depression (Kessler et al., 2005a; Sweet et al., 1992). In particular, medical students present high prevalence rates of depression (Goldman et al., 2015).

Our hypothesis was that CPT measures were indicative of primary inattention deficits and not secondary to depressive symptomatology. We tested this hypothesis investigating the performance on CPT by a sample of young adults controlling for both ADHD and depression symptoms in a dimensional way (number of symptoms), which might provide more reliable data (Larsson et al., 2012). For the purposes of this study, Major Depression was an exclusionary criterion.

2. Methods

2.1. Subjects

The initial sample consisted of 143 medical students from a public university in Rio de Janeiro, Brazil, which volunteered for this study. Ninety-three students completed protocol while 50 had some of the main variables of interest (ADHD, depressive symptoms or CPT) incomplete (2 students missed all three variables and 52 students had any combination of 2 variables missing). There was no statistical significant difference between the socio demographic profile of completers and non-completers. The complete sample consisted of 60 (64.5%) women and 33 (35.5%) men with a mean age of 24 years old.

2.2. Procedures

Evaluation of all participants consisted of the ASRS - Adult Self Rating Scale (Mattos et al., 2006) and a semi-structured interview using DSM-IV criteria (K-SADS for ADHD (Kaufman et al., 1997)) by board certified psychiatrists with more than 10 years of experience in adult ADHD (BPN and PM); diagnoses were made upon discussion of cases (history, self-report and semi-structured interview). Since there is some evidence that self-report of ADHD symptoms may be somewhat unreliable (Sibley et al., 2012) ADHD symptomatology was assessed not only by self-report (ASRS) but also the semi-structured interview (K-SADS). All ADHD assessments were done blind to the screening status and all students with more than 5 current symptoms of inattention or hyperactivity/impulsivity were discussed with the other rater. Participants completed the Beck Depression Inventory (BDI) (Beck and Beamesderfer, 1974). Individuals with current Depressive Episode were not included for the purposes of this study; other exclusion criteria were psychosis, epilepsy and current use of benzodiazepines or antidepressants. All participants signed an informed consent; the local Ethics Committee approved this study.

Neuropsychological evaluation comprised CPT and IQ, calculated with the Four-Subtest Form of the Wechsler Abbreviated Intelligence Scale (WASI) (Beck et al., 2009), from which the blocks, vocabulary, matrix and similarities were administered. The Conner's Continuous Performance Task II (CPT) (Conners, 2004) is the most commonly used attention task in clinical practice (Rabin, Barr, & Burton, 2005) giving the opportunity to evaluate the ability of a subject to maintain consistent responding (Robertson, Manly, Andrade, Baddeley, & Yiend, 1997) it primarily measures sustained attention, distractibility and response inhibition through errors (omissions and commissions), reaction time (RT) and RT variability. Variables of interest in the present study were: Number of Omission errors (OMI), Commission errors (COM), Hit Reaction Time (HRT), HRT Block Change (HRT BL CHANGE), Reaction Time by Inter-Stimulus Interval (Hit RT ISI Change) and Attentiveness (d'). OMI occur when subjects fail to respond on trials containing target letters (all non-"X" letters), COM occur when they respond on trials with letters "X". HRT is the mean response time for all non-X responses over all six-time blocks and represents the subject's easier discrimination of the target. HRT BL CHANGE (a vigilance measure) is the slope of change in the reaction time over the six time blocks; a positive slope indicates a slowing RT and a negative slope indicates quicker RT as the test progresses. Hit RT ISI CHANGE (capacity to adjust to presentation speed) is calculated by computing the slope of change in RT over the three ISIs (1, 2, and 4 s). The ISIs are block-randomized so that all three ISI conditions would occur every block but in a dif-

ferent order; by varying the inter-stimulus intervals (1, 2 and 4 s) it is possible to assess the ability to adjust to changing tempo and task demand. The index d' reflects the subject's perceptual sensitivity to targets, in other words, it is a measure of how well the individual discriminates between targets (signals) from non-targets (noise). Higher d' values indicate greater sensitivity and better discrimination between targets and non-targets.

2.3. Statistical analysis

First, we investigated the correlation between depression and ADHD symptoms, either measured by self-report (ASRS) or semi-structured interview (K-SADS). Statistical tests were then performed to verify the significance of the correlation of CPT scores with the number of inattentive and hyperactive symptoms (defined by K-SADS scores) and the depressive symptoms (defined by BDI total scores). A K-SADS composite score was calculated with the total number of symptoms (inattention and hyperactivity/impulsivity); a second composite score was calculated using this first composite score multiplied by BDI score. Except for d' , all distributions of the scores were not normal, and Spearman correlation was used in the analysis. Correlations (Spearman) were made among OMI, COM, d' , HRT, HRT BL and HRT ISI CHANGE results and the BDI, ASRS and KSADS values in order to identify initially the association of the studied variables. Afterwards, we performed multiple linear regressions to test our hypothesis, considering the CPT measures as the dependent variables in each regression, and KSADS composite score, BDI score, and KSADS Comp $\times$ BDI, as independent variables. These scores were natural log transformed prior to statistical modeling to approximate a normal distribution. A probability level of $p < 0.05$ was used; correlations above 0.3 were considered significant.

3. Results

The sample had a mean IQ of 115 (high average). There was no significant interaction between IQ, gender, ADHD symptoms and CPT measures. Regarding ADHD symptoms, 31% had at least 6 current inattention symptoms and 9% had at least 6 current hyperactivity-impulsivity symptoms; formal diagnosis using DSM-IV criteria in this sample was 4.57%.

Depression was absent in 72.5% of the sample; 17.5% had mild depressive symptoms and 10% had moderate depressive symptoms according to the BDI classification. There was no correlation between depression and ADHD symptoms, either measured by self-report (ASRS scores) or semi-structured interview. Depression was not significantly correlated either with inattention ($\rho = 0.12$) or hyperactivity-impulsivity ($\rho = -0.03$). Also, additional analysis demonstrated that ASRS items were not correlated with the BDI total score.

Multiple regression analysis is presented in Table 1. The only variable that significantly predicted changes in CPT was the KSADS Comp Score, with a standardized B of 0.186 for CPT Omission; 0.291 for CPT Commission and -0.226 for CPT d' . The scatterplot of the CPT Omission, CPT Commission and CPT d' Prime by KSADS CS prediction is given, respectively, in Figs. 1–3.

4. Discussion

This study aimed to investigate the influence of depressive and ADHD symptoms on attention performance using one of the most common tasks in clinical practice, the CPT. Both ADHD and depression can be measured dimensionally and comprise inattention as part of the clinical picture. Individuals presenting both depressive symp-

Table 1

Multiple linear regressions with CPT factors as independent variables.

	R ²	Adjusted R ²	Standardized B	Significance
CPT Omission				
KSADS CS	0.35	0.025	0.186	<0.05
BDI	0.35	0.17	0.025	N. S.
KSADSCS $\times$ BDI	0.35	0.07	-0.011	N. S.
CPT Commission				
KSADS CS	0.085	0.076	0.291	<0.05
BDI	0.105	0.088	0.144	N. S.
KSADSCS $\times$ BDI	0.106	0.8	-0.65	N. S.
CPT HRT				
KSADS CS	0.001	-0.011	-0.028	N. S.
BDI	0.016	-0.007	-0.124	N. S.
KSADSCS $\times$ BDI	0.016	-0.019	0.024	N. S.
CPT St Error				
KSADS CS	0.001	-0.01	-0.038	N. S.
BDI	0.011	-0.13	-0.098	N. S.
KSADSCS $\times$ BDI	0.013	-0.022	-0.106	N. S.
CPT D Prime				
KSADS CS	0.051	0.04	-0.226	<0.05
BDI	0.078	0.56	-0.0164	N. S.
KSADSCS $\times$ BDI	0.087	0.054	0.197	N. S.
CPT HRT BL Change				
KSADS CS	0.028	0.016	-0.166	N. S.
BDI	0.046	0.023	0.135	N. S.
KSADSCS $\times$ BDI	0.013	-0.022	-0.106	N. S.
CPT HRT ISI Change				
KSADS CS	0.002	-0.009	0.047	N. S.
BDI	0.005	-0.019	-0.052	N. S.
KSADSCS $\times$ BDI	0.011	-0.024	0.170	N. S.

$p < 0.05$; N.S. = Non-Significant; KSADS CS = KSADS Composite Score (Inattention+Hyperactivity/Impulsivity symptoms); BDI = Beck Depression Inventory; KSADSCS $\times$ BDI = KSADS Composite Score $\times$ Beck Depression Inventory

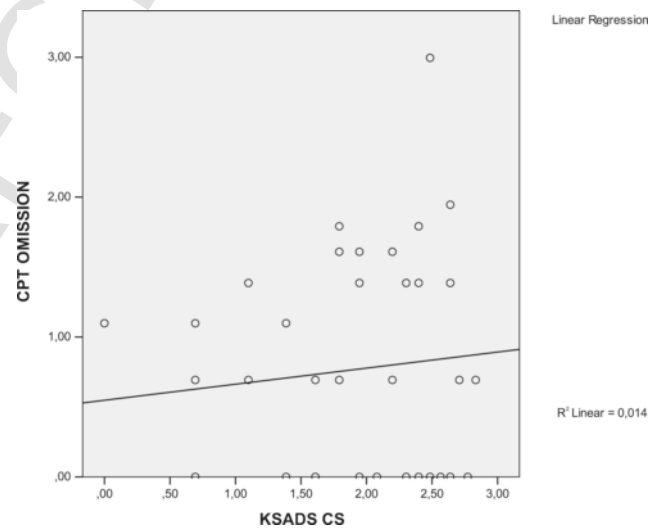


Fig. 1. Linear regression scatterplot CPT omission (dependent variable) And KSADS composite score (independent variable).

toms and inattention are often referred to neuropsychological evaluation, where CPT is commonly considered the main task addressing attentional skills. Of note, regularly the patient's referral has different diagnostic purposes. In this study, ADHD symptoms were neither correlated with nor predicted by BDI scores suggesting that an increasing number of ADHD symptoms were not associated with more depressive symptoms. Also, depression was not correlated with any of the CPT scores suggesting a lack of impact on attentional performance in this sample. We suggest that ADHD associated impairment on the CPT is not augmented by the presence of mild depressive

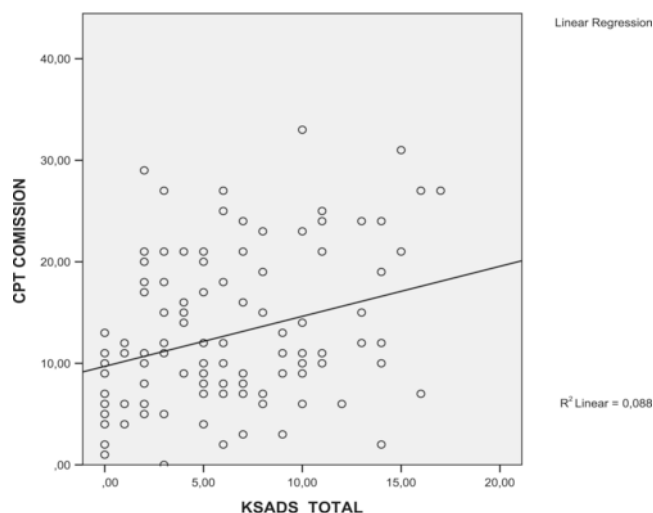


Fig. 2. : Linear regression scatterplot CPT commission (dependent variable) and KSADS composite score (independent variable).

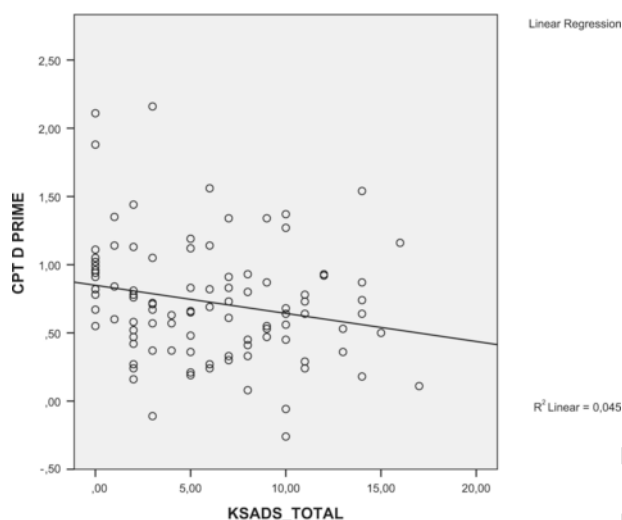


Fig. 3. : Linear regression scatterplot CPT D prime (dependent variable) and KSADS_TOTAL = composite score (independent variable).

symptoms because of a ceiling effect. Furthermore, mild depressive symptoms do not add extra impairment to that which is already present and associated with ADHD symptoms.

In our sample, OMI and COM, as well as d' were correlated to ADHD symptoms (K-SADS composite score); the percentage of prediction for COM was higher than for OMI. An impaired capacity to inhibit a pre-potent motoric response (Barkley, 1997) could explain higher COM errors (Rivenes et al., 2009) and a lack of effort necessary to meet task demands (Luman et al., 2009) could explain both OM and COM errors in CPT, even when primary monitoring functions are intact (Barkley, 1997). Also, ADHD symptomatology was the only predictor of changes in those CPT variables and such measures may help in the differential diagnosis (Depression only versus Depression comorbid with ADHD). These findings are in accordance to those of the only other somewhat similar study in our country (Malloy-Diniz et al., 2007).

There are several theoretical models explaining neuropsychological deficits in ADHD, which could explain higher OMI and COM errors in CPT. Using a different methodology, our results are in accor-

dance with Hill et al., (2008), who suggested that when attention deficits are observed they are more likely not accounted for depression. This finding is particularly helpful for the neuropsychologist who must conclude on the nature of attention deficits when both ADHD and depressive symptoms are present. Whenever depression is considered to cause inattention, clinicians tend to either postpone or discard ADHD diagnosis and invest in depression treatment. If ADHD symptoms were considered as the source of attention complaints, clinicians would consider an ADHD diagnosis in the first place, even if depressive symptoms merited a parallel treatment. Our results suggest that CPT measures demonstrating inattention correlates to ADHD symptomatology even if another potential contributing factor, as depression is present.

5. Limitations

It should be noted that individuals with major depression were not included for the purposes of this study; our sample mirrors the proportion of depression described in similar studies (Murphy et al., 2002), we intended to evaluate individuals with milder depressive symptomatology whose neuropsychological testing results must be interpreted either as indicative of a primary attention deficit or secondary to depression.

Additional analyses of potential contributing factors for CPT performance, which are beyond the scope of this study, like executive functions measures, should also be investigated in future studies. We have evaluated a sample of young university students and it should be noted that our results might not be generalized to other different samples.

Conflicts of interest

Paulo Mattos was on the speakers' bureau and/or acted as consultant for Janssen-Cilag, Novartis, and Shire in the previous years; he received travel awards to participate in scientific meetings from those companies. The ADHD outpatient program (Grupo de Estudos do Déficit de Atenção/Institute of Psychiatry) chaired by Dr. Mattos has also received research support from Novartis and Shire.

Joseph A. Sergeant has served on the advisory boards of Lilly and Shire, and received grants from Lilly. He has also received speaker's fees from Shire, Lilly, Janssen-Cilag, and Novartis. The other authors do not state any conflict of interests.

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